

SBLToolkit.jl & Generalized Similarity Profile Theory (GSPT) Core Math Cheatsheet

Compiled for Advanced Boundary-Layer Research and Single-Column Modeling (SCM) Audits

Transforms physical vertical profiles into flux-coordinate mapping spaces to decouple coordinate stretching from physical collapses.

- **Similarity Mapping Coordinate** $\zeta(z)$:

$$\zeta(z) = z/L(z) \quad (L(z) \text{ is Obukhov length profile})$$

- **Coordinate Jacobian** $\zeta_z \equiv \frac{\partial \zeta}{\partial z}$:

$$\zeta_z = \frac{L(z) - zL'(z)}{L(z)^2} = \frac{1 - \zeta L'(z)}{L(z)}$$

- **Coordinate Curvature** $\zeta_{zz} \equiv \frac{\partial^2 \zeta}{\partial z^2}$:

$$\begin{aligned} \zeta_{zz} &= -\frac{zL''(z)}{L(z)^2} - \frac{2L'(z)}{L(z)} \zeta_z \\ &= -\frac{L''(z)}{L(z)} \zeta - \frac{2L'(z)}{L(z)} \zeta_z \end{aligned}$$

- **Height-Resolved Curvature Decomposition:**

$$Ri_{zz} \equiv \frac{d^2 Ri_g}{dz^2} = S'_h(\zeta) \zeta_z^2 + S_m(\zeta) \zeta_{zz} + \mathcal{E}_{\Delta z}$$

- **Linear Obukhov Profile Test** (No Fold Gate): If vertical flux divergence is linear ($L(z) = L_0 + az$):

$$L''(z) = 0 \implies \zeta_z = \frac{L_0}{(L_0 + az)^2} > 0 \quad \forall z \geq 0$$

No finite coordinate folds can exist; mapping is globally monotone.

- **Nondegenerate Coordinate Fold Condition:**

$$\zeta_z(z^*) = 0 \iff L(z^*) = z^* L'(z^*) \quad \text{under } L''(z^*) > 0$$

At the fold locus z^* , observed curvature is purely coordinate-driven:

$$\begin{aligned} Ri_{zz}(z^*) &= S_m(\zeta^*) \zeta_{zz}(z^*) \\ &= -\frac{z^* L''(z^*)}{L(z^*)^2 (1 + \beta_m \zeta^*)^2} \end{aligned}$$

Models stable boundary layers as coupled singular perturbation systems in 2D phase space.

- **Fast Subsystem TKE Equation** ($\dot{e}$, $e \geq 0$):

$$\epsilon \frac{de}{dt} = \mathcal{P} - \mathcal{B}(e) - \epsilon_{\text{diss}}$$

$$\epsilon \frac{de}{dt} = l_0 \sqrt{e + \delta} S^2 - B_{0,\text{max}} \left(\frac{e^2}{e^2 + \delta_{\text{reg}}^2} \right) - \frac{e^{3/2}}{l_0}$$

- **Velocity Regularization Floor** δ : Set to $10^{-6} \text{ m}^2 \text{ s}^{-2}$, restoring C^1 -continuity of the vector field across zero-energy limits.

- **Slow Subsystem Wind Shear Equation** ($\dot{S}$, $S \geq 0$):

$$\frac{dS}{dt} = G_0 - \gamma_s e S - r_s S$$

- **Fast Subsystem Stability Eigenvalue** (λ_f):

$$\lambda_f \equiv \frac{l_0 S^2}{2\sqrt{e + \delta}} - \frac{2B_{0,\text{max}} e \delta_{\text{reg}}^2}{(e^2 + \delta_{\text{reg}}^2)^2} - \frac{3\sqrt{e}}{2l_0}$$

- **Unregularized Saddle-Node Fold Locus** ($\lambda_f = 0$):

$$x^3 + ax^2 - C = 0$$

$$\text{where } x \equiv S_{\text{fold}}^2, a \equiv \beta N^2, C \equiv \frac{27B_{0,\text{max}}^2}{4l_0^4}$$

$$e_{\text{fold}} = \frac{l_0}{\sqrt{3}} \sqrt{S_{\text{fold}}^2 + \beta N^2} = \frac{3B_{0,\text{max}}}{2l_0 S_{\text{fold}}^2}$$

- **Full 2D System Saddle-Node Boundary** ($\det J = 0$):

$$\det J = \frac{1}{\epsilon} [-F_e(\gamma_s e + r_s) + \gamma_s S F_S] = 0$$

- **Manifold Adhesion Distance** ($d_{\perp}$):

$$d_{\perp}(t) = \frac{|F(e, S)|}{\|\nabla_{e,S} F\| + \epsilon_{\text{floor}}}$$

Analytically couples prognostic land-surface energy balance directly into vertical tridiagonal diffusion arrays to eliminate stiff numerical instabilities.

- **Continuous Prognostic Surface Energy Balance:**

$$C_s \frac{\partial T_s}{\partial t} = R_n(t) - H_0(t) - G_s(t)$$

- **Boundary Sensible Heat & Ground Conduction Fluxes:**

$$H_0 = \rho c_p K_h(z_1) \left(\frac{\theta_1 - T_s}{z_1} \right), \quad G_s = K_s (T_s - T_{\text{deep}})$$

- **Simultaneous Backward-Euler Discretization:**

$$\gamma \equiv \frac{\rho c_p K_h^{n+1}}{z_1}, \quad \beta \equiv \frac{C_s}{\Delta t} + \gamma + K_s$$

- **Analytical Elimination of surface** T_s^{n+1} :

$$T_s^{n+1} = \frac{1}{\beta} \left[\frac{C_s}{\Delta t} T_s^n + R_n^{n+1} + \gamma \theta_1^{n+1} + K_s T_{\text{deep}} \right]$$

- **Tridiagonal Atmospheric Row Integration** (z_1):

$$\frac{\theta_1^{n+1} - \theta_1^n}{\Delta t} = \frac{K_h^{n+1}}{\Delta z_1} \left[\frac{\theta_2^{n+1} - \theta_1^{n+1}}{\Delta z_{12}} - \frac{\theta_1^{n+1} - T_s^{n+1}}{z_1} \right]$$

Substituting T_s^{n+1} absorbs the boundary term into the primary diagonal (B_1) and superdiagonal (C_1), solved in-place.

SBL Inversion Standards, Empirical Closures, and Campaign Reference

Auxiliary Standard Scaling Formulations, GSPT Taxonomy, and Operational NWP Mandates

Smooths linear, Gaussian sensor fields prior to taking non-linear spatial quotients to bypass the non-commutation hazard.

- **The Non-Commutation Hazard:**

$$\mathcal{M}_\delta \left[\frac{A}{B} \right] \neq \frac{\mathcal{M}_\delta A}{\mathcal{M}_\delta B}$$

Smoothing Ri_g artificially generates knees.

- **Wavenumber Transformation Coordinate:**

$\xi \equiv \ln(z/z_0)$ where z_0 is aerodynamic roughness

- **Tikhonov-Morozov Regularization Spline:**

$$\min_{f_s} \sum_{i=1}^{N_{\text{obs}}} \left(\frac{f_s(\xi_i) - q_i}{\sigma_i} \right)^2 + \alpha \int_0^{\xi_{\text{max}}} [f_s''(\xi)]^2 d\xi$$

- **Morozov's Discrepancy Principle (MDP):** Solve for α via bisection until: $\|f_s(\xi_i) - q_i\|_2 = \delta$.

- **Tangential Cone Condition** (γ non-linearity bound):

$$\|F(x_2) - F(x_1) - F'(x_1)(x_2 - x_1)\|_Y \leq \gamma \|F(x_2) - F(x_1)\|_Y$$

Guarantees stable numerical convergence of bisection sweeps.

- **Analytical Log-Height Derivatives:**

$$\frac{df}{dz} = \frac{1}{z} \frac{df}{d\xi}, \quad \frac{d^2f}{dz^2} = \frac{1}{z^2} \left(\frac{d^2f}{d\xi^2} - \frac{df}{d\xi} \right)$$

SBL stability functions (S_m, S_h) expressed directly in terms of the gradient Richardson number (Ri).

- **Prandtl-Normalized Symmetric SBL (for $Pr = 1$):**

$$S_m(Ri) = S_h(Ri) = \begin{cases} \left(\frac{Ri_c - Ri}{Ri_c} \right)^2, & 0 \leq Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases}$$

* *Note: The quadratic form preserves continuous vertical derivatives at $Ri_c = 0.25$, eliminating numerical chatter aloft.*

- **GLGS (eMOST / Modified Grachev et al. 2007):**

$$S_m(\zeta) = \phi_m^{-2} = \left[1 + \frac{a_m \zeta}{(1 + b_m \zeta)^{2/3}} \right]^{-2}$$

$$S_h(\zeta) = \left[\phi_m \cdot Pr_0 \left(1 + \frac{a_h \zeta}{1 + b_h \zeta} \right) \right]^{-1}$$

* *Optimized constants: $Pr_0 = 0.98, a_m = a_h = 5.0, b_m = 0.3, b_h = 0.4$.*

- **Integrated Stability Correction Functions:**

$$\psi_m(\zeta) = -3 \frac{a_m}{b_m} \left[(1 + b_m \zeta)^{1/3} - 1 \right]$$

$$\psi_h(\zeta) = -Pr_0 \frac{a_h}{b_h} \ln(1 + b_h \zeta)$$

- **Businger-Dyer (Log-Linear Limit):**

$$\phi_m(\zeta) = 1 + \beta_m \zeta, \quad \phi_h(\zeta) = Pr_0(1 + \beta_h \zeta) \quad (\zeta < 0.1)$$

SBL Quality Control & Verification Metrics

- **Triple-Point Dispersion (δ_{TP}):**

$$\delta_{TP}(t) = \frac{\max(z_K, z_e, z_{e_z}) - \min(z_K, z_e, z_{e_z})}{H_{SBL}(t)} \leq 0.10$$

* *Tracks cutoff convergence. Physical collapses are gated where $\delta_{TP} < 0.1$.*

- **Wave-Jet CEOF Separation Phase Gra-**

Standards for vertical tower footprints and multi-campaign diagnostics.

Campaign	N_z	Dominant Geometry	GSPT Inv.
CASES-99	150	Jet nose fold ($\zeta_z = 0$)	Success
GABLS3	150	Shear erosion/cooling	Success
SHEBA	2	Stencil collapse ($N_z < 3$)	Bypassed

The Four-Level Diagnostic Hierarchy:

Coordinate Fold $\implies$ Cubic Ri Degeneracy $\implies$ Loss of Fast Hyperbolicity $\implies$ Saddle-Node

The Four SCM Operational NWP Mandates:

1. **Apply Track A Pre-Diagnostic Smoothing:** Filters primitive velocity and temperature fields (u, v, θ_v) in log-height space before differentiating to prevent S^{-6} variance blowup aloft.
2. **Implement Fast Stability Gating (λ_f):** Shut-off vertical mixing only if $\lambda_f \geq -\epsilon_\lambda$ is met, preventing coordinate fold illusions ($\zeta_z = 0$) from cutting off fluxes.
3. **Cease Empirical Parameter Tuning:** Do not retune Monin-Obukhov constants (β_m, β_h) to fit physical profile knees; these knees are coordinate mapping features.
4. **Enforce $N_z \geq 3$ Vertical Ingestion Gates:** Bypasses finite difference curvature operators (**D**₂) on sparse grids to prevent stencil collapse. Enforce Bulk Richardson fallbacks:

$$Ri_b = \frac{g}{\theta_v} \frac{(z_2 - z_1)(\theta_{v,2} - \theta_{v,1})}{(u_2 - u_1)^2 + (v_2 - v_1)^2}$$

SBLToolkit.jl Open Source Core - Technical Reference Sheet